

Reading a Single-Line Diagram

Category: Panels & Distribution

This primer explains how to interpret a typical LV single-line diagram in a project drawing set.

1. Identify the supply source (utility transformer or upstream panel) and note the rated voltage, short-circuit level and earthing.
2. Follow the main incomer into the main switchboard, locating the main breaker and any metering or surge protection.
3. Trace each outgoing feeder to downstream distribution boards, noting breaker ratings, cable sizes and routing references.
4. For each final circuit, understand what the circuit serves (lighting, sockets, HVAC, EV, etc.) and how it is protected.
5. Use circuit identifiers (e.g. L1-01, L2-05) consistently when labelling, commissioning and fault finding.

Use this tutorial together with the project-specific as-built drawings and applicable standards; do not rely on it as a complete d